

Medical Imaging Dataset Exploration Explore any two medical imaging datasets and collect the following information:

1. Type of imaging data (X-ray, CT, skin, etc)
2. Number of images
3. Available classes or labels
4. Dataset imbalance
5. Any challenges you observe (noise, annotation quality, etc.)

Dataset #1

Xray data set from Kaggle - <https://www.kaggle.com/datasets/alifrahman/covid19-chest-xray-image-dataset?resource=download>

Type of imaging data: Chest X-Rays (Radiography).

Number of images: 94 total images.

Available classes: Two labels — covid and normal.

Dataset imbalance: Severe imbalance. There are 69 COVID-19 images (~73%) and only 25 Normal images (~27%).

Challenge 1 - Overfitting Risk: 94 images is an incredibly tiny dataset for deep learning. The AI will likely memorize the specific shapes in these 94 images rather than learning real medical features, meaning it will fail on new patients.

Challenge 2 - Skewed Accuracy: Because of the imbalance, an AI could achieve 73% accuracy just by blindly guessing "COVID" on every single image without actually looking at the X-ray.

Challenge 3 - Lack of Metadata: There is no patient history, age, gender, or severity information provided, which makes clinical validation impossible.

Dataset#2 Mouse Cranial Bone Micro-CT Scans (by chzpan)

<https://www.kaggle.com/datasets/chzpan/bone-lab>

Type of imaging data: Micro-CT Scans (Computed Tomography). These are 2D cross-sectional slices of 3D bone structures, saved as greyscale PNGs.

Number of images: ~5,606 slice images (taken from 65 distinct bone samples).

Available classes or labels: Instead of text

categories (like "Sick" or "Healthy"), the labels here are pixel coordinates (Regions of Interest / ROI). The ground truth data provides the exact X and Y coordinates of a surgically created circular defect in the rat's cranial bone.

Dataset imbalance: Because this is not a classification task, traditional "class imbalance" does not apply. However, there is a severe spatial imbalance: the actual bone defect we are trying to detect is incredibly tiny compared to the massive, empty black background of the 2048x2048 pixel image.

Challenge 1 - Complex Label Parsing: Unlike the COVID dataset where images were just neatly sorted into two folders, the "labels" here are raw text files containing coordinates. We would have to write custom Python code to read the text file, extract the numbers, and mathematically map them to the correct image.

Challenge 2 - Massive Resolution: The images are natively 2048x2048 pixels. Feeding images that large into a standard neural network will likely cause our computer to crash from lack of RAM. We need to heavily resize or crop them, which introduces the risk of accidentally blurring or cutting out the tiny bone defect

Challenge 3 - Loss of 3D Context: CT scans are inherently 3D structures. By splitting them into flat 2D slice images for the AI to analyze one by one, the model loses the 3D depth context of how the bone connects together.